Chenghao Wu

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theo-wu.github.io

Education

University of Manchester — **Manchester, UK**
PhD in Computer Science (Supervisor: Dr. Zahra Montazeri) — *March 2023 – present*
Focus: differentiable rendering, appearance reconstruction, neural rendering

University College London — **London, UK**
MSc in Computer Graphics, Vision and Imaging — *Sept 2021 – Sept 2022*

Sun Yat-sen University — **Canton, China**
BEng in Intelligence Science and Technology — *Sept 2017 – June 2021*

Publications and Manuscripts

Appearance and Geometry Reconstruction of Woven Cloth from a Single Close-up Image
In submission
Chenghao Wu, *Apoorv Khattar, Ling-Qi Yan and Zahra Montazeri*

Learning to Rasterize Differentiably
Computer Graphics Forum (EGSR 2024)
Chenghao Wu, *Hamila Mailee, Zahra Montazeri and Tobias Ritschel*

An End-to-End ConvLSTM-based Method for Point Cloud Streaming Compression
International Conference on Advanced Robotics and Mechatronics 2024
Junhua Long, Mingjian Feng, Boyang Li, Yehua Ling, ***Chenghao Wu****, Kai Huang and Mingyue Cui*

Cooperative LiDAR SLAM for multi-vehicles based on edge computing
Journal of Image and Graphics 2021
Mingyue Cui, Shipeng Zhong, Siyao Liu, Boyang Li, ***Chenghao Wu*** *and Kai Huang*

Internships

Jiangxi Kmax Industrial Co., Ltd — **Shenzhen, China**
Graphics Engineer Intern — *June 2021 – Sept 2021*
- Contributed to skin reconstruction for virtual avatars.

Personal Projects

MetaRas: Meta-Learned Differentiable Rasterizer — **May 2022 – Nov 2022**
- Two implementations: one in JAX with auto differentiation, one in Pytorch with C++/CUDA extensions.
- Supports texturing, shading, provides several simple tasks, and integrates meta-learning into rasterization.

Scholarships

Computer Science Department Scholarship — **University of Manchester**
Funds three years of PhD studies — *April 2023 – April 2026*

Outstanding Student Scholarship — **Sun Yat-sen University**
Awarded for excellent academic performance in sophomore year (10%) — *September 2019*

Skills

Programming: C++, Python, GLSL, LaTeX, C#
Applications: CUDA, JAX, Mitsuba, Pytorch, Unity, WebGL
Languages: Mandarin (native), Cantonese (native), English (working)
Interests: Sport, Movie, Sci-Fi